COURSE DETAILS

1. Course title: Multimedia Communications
2. Course code: CSE 4295
3. Credit hours: **3**
4. Level: **Year: 4, Semester: 2**
5. Semester offered: **Spring 2025**
6. Programme: **B.Sc. in Electrical and Electronic Engineering (EEE)**
7. Department: **Electrical and Electronic Engineering (EEE)**
8. Faculty: **Engineering**
9. Synopsis:

Introduction: Types of media; Multimedia signal characteristic: sampling, digital representation, signal formats; Signal coding and compression: entropy coding, transform coding, vector quantization; Coding Standards: H.26x, LPEG, MPEG; Multimedia communication networks: network topologies and layers, LAN, MAN, WAN, PSTN, ISDN, ATM, internetworking devices; the internet and access technologies; enterprise networks; wireless LANs and wireless multimedia; Entertainment networks: cables, satellite and terrestrial TV networks; ADSL and VDSL; high speed modems; Transport protocols: TCP, UDP, IP, Ipv4, Ipv6, FTP, RTP, and RTCP; use of MPLS and WDMA; Multimedia synchronization; security; QoS and resource management; Multimedia applications: The www; Internet telephony; teleconferencing; HDVT; e-mail and e-commerce.

10. Type of course (core/elective): **Elective**
11. Prerequisite(s) (if any): N/A
12. Name of the instructor(s) with contact details and office hours:

Course Instructor's Name:	Faisal Farhan
Room:	9C Teachers Room
Phone:	01778726056
E-mail:	faisalfarhan.eee@aust.edu
Consultation Hour:	• Sunday 2.40 p.m - 3.30 pm • Monday 11.20 a.m-1.00 p.m • Thursday 1.50 p.m - 3.30 p.m

13. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Bloom's Taxonomy Level

After completion of the course, the students will be expected to:

CO No.	CO statement	PO No.	K/P /A	Bloom's Taxonomy (Domain/ level)	Delivery Methods and Activities	Assessment Method
1	Explain the working principles of multimedia signal characteristics, modern multimedia networks, transport protocols as well as contemporary techniques for encoding, compressing and providing security to various multimedia data types.	PO (a)	K4/ P1, P2, P3	Cognitive/ Understand	- Lecture - PP Presentation - Video Demonstration - Brain Storming Session - Think – Pair -Share (TPS)	- Class Performance - Quiz - Final Exam
2	Comprehend the various aspects of different multimedia networks, transport protocols, encoding techniques, data compression techniques and security measures.	PO (a)	,K4/ P1, P2, P3	Cognitive/ Understand	- Lecture - PP Presentation - Video Demonstration - Brain Storming Session - Think – Pair -Share (TPS)	- Class Performance - Quiz - Final Exam
3	Apply existing techniques to encode, compress as well as secure different types of multimedia data.	PO (b)	K4/ P1, P2, P4	Cognitive/ Apply,	- Lecture - PP Presentation - Video Demonstration - Brain Storming Session - Think – Pair -Share (TPS)	- Class Performance - Quiz - Final Exam

Description of PO Indicators

PO Description
(a)Engineering knowledge: Apply knowledge of mathematics, natural science, (*electrical and electronic / electronic and communication)* engineering fundamentals, and specialization to the solution of complex (*electrical and electronic / electronic and communication)* engineering problems (CEP). (K1 to K4)
(b) Problem analysis: Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4)
(c) Design of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)

Description of Applicable Knowledge Profiles (K3,K4,K5) and CEP Attributes (P1, P2)

K3	**Engineering fundamentals**	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline

K4	**Engineering specialization**	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5	**Design**	Knowledge that supports engineering design in a practice area
P1	**Depth of knowledge required**	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach.
P2	**Range of conflicting requirements**	Involve wide-ranging or conflicting technical, engineering and other issues.

14. Percentages of Assessment Methods

Method	Percentage
Class Performance	**10**
Quizzes/Assignments	**20**
Final Examination	**70**

15. Week wise distribution of contents and assessment methods

Week	Topics	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
1	Introduction to Multimedia Types of Multimedia	Lecture	- Class Performance - Quiz 1	01
2	Multimedia Digitization Principles Digital Goods Digital Communication and Data Transmission Multipoint Conferencing Network and Application QoS The Advance of Personal Telecommunications			
3	Network Topologies Data Network Network Types Telephone and Service Network Broadcast TV Network Broadband Multiservice Internet & Access Technologies	Lecture	- Class Performance - Quiz 1	
4	Multimedia Networks Communication Fundamentals in Computer Networks Basic Terms and Concepts	Lecture	- Class Performance - Quiz 1	
5	Computer Networks and Network Protocols	Lecture	- Class Performance - Quiz 2 - Final Exam	01 and 02
6	Performance Ratios of Computer Networks	Lecture	- Class Performance - Quiz 2 - Final Exam	01 and 02

Week	Topics	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
7	Error Detection and Correction in Computer Networks	Lecture	- Class Performance - Quiz 2 - Final Exam	01 and 02
8	Multimedia Data and Its Encoding Media Variety and Multimedia	Lecture	- Class Performance - Quiz 2 - Final Exam	01 and 03
9	Text – Data Formats and Compression Graphics – Data Formats and Compression	Lecture	- Class Performance - Quiz 3/Assignment - Final Exam	01 and 03
10	Audio – Data Formats and Compression	Lecture	- Class Performance - Quiz 3/Assignment - Final Exam	03
11	Video and Animation – Data Formats and Compression	Lecture	- Class Performance - Quiz 3/Assignment - Final Exam	
12	Digital Security	Lecture	- Class Performance - Quiz 3/Assignment - Final Exam	03
13	Principles of Security in Computer Networks Confidentiality and Encryption	Lecture	- Class Performance - Quiz 4/Assignment	
14	Digital Signatures Public Key Infrastructures and Certificates	Lecture	- Class Performance - Quiz 4/Assignment	03

16. References

16.1. Required

1. *"Digital Communication - Communication, Multimedia, Security"*, Christoph Meinel and Harald Sack, (Springer).

16.2. Recommended

1. *Multimedia Communications: Applications, Networks, Protocols And Standards"*, Fred Halsall, (Pearson Education).
2. *"Multimedia Communications and Networking"*, Mário Marques da Silva, (CRC Press).
3. *"Multimedia Communications: Directions and Innovations"*, Jerry D. Gibson, (Academic Press).
4. Lecture notes. (Available at *http://adnan.quaium.com/aust/cse4295*)
5. Lecture notes. (Available at *https://classroom.google.com/c/ODMxNDc3MTExMzM1)*

Prepared by:	Checked by:	Approved by:
Signature: _____ Name: Faisal Farhan Department: **EEE** Date:	Signature: _____ Name: **Dr. Shah Md. Salimullah** OBE Program Coordinator, **EEE** Date:	Signature: _____ Name: **Prof. Dr. Bobby Barua** HoD, **EEE** Date: